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Turbulence

Assignment 4 (CFD)

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1 Introduction

The goal of this task is to comprehend the basics of turbulence, including the energy cascade, inter-scale interaction, and the behavior of convective and diffusive terms in the kinetic energy transport equation.

The Navier-Stokes equations are the most extensive equations that govern the behavior of fluids, including turbulence. Nevertheless, solving these equations is complex and necessitates the use of numerical techniques. Using these techniques, the task of solving the equations is still laborious. Therefore, simplifications are typically employed to achieve the outcome of these equations. In this task, the Burgers equation is utilized to examine the kinetic energy in a turbulent fluid.

1.1 Problem specification

This assignment involves solving Burgers equations in Fourier space through direct numerical simulation (DNS), where all turbulence scales in space and time are resolved directly. While this method provides very precise outcomes, it comes with a substantial computational expense. Turbulence models are used to simplify the issue and lower the computational expense. For example, large eddy simulation (LES) specifically targets larger motion scales, further simplifying the issue. This task also addresses the Burgers equation by employing this simulation method.

As a result, the Burger's equation will be addressed through DNS and LES methods, followed by analysis and comparison of the outcomes. Furthermore, a short parametric analysis will be carried out to explore the behavior of the solution with different inputs.

2 Methodology

Non-linear dynamics of turbulent flow is represented by the Navier-Stokes equations. Its dimensionless equations for incompressible flow stands as:

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)\, u = \frac{1}{Re}\Delta u - \nabla p \;\; ; \qquad \nabla u = 0 \tag{1}$$

Direct numerical simulations (DNS) of the Navier-Stokes (NS) equations require significant computational resources because the convective term generates multiple dynamically important scales of motion. In contrast, Burgers' equation provides a simplified framework that retains numerous characteristics of the NS equations:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \frac{1}{Re}\frac{\partial^2 u}{\partial^2 x} + f \tag{2}$$

The Burgers' equation in Fourier space offers a manageable framework for examining turbulence behavior. The following equation (3), obtained from the Navier-Stokes equations, balances convective and diffusive components. In this context, u_k denotes the Fourier transform applied to the velocity field, ν symbolizes the kinematic viscosity, and F_k refers to any external forcing acting on the system. The convective term aids in energy movement across scales, whereas the diffusive term diminishes energy, especially at smaller scales.

$$\partial_t \hat{u_k} + \sum_{k=p+q} \hat{u_p} i q \hat{u_q} = -k^2 \nu \hat{u_k} + F_k \quad k = 0,\ 1,\ 2,\ ...,\ N \tag{3}$$

A clear time-integration method is used to solve Burgers' equation, guaranteeing computational stability and adhering to the Courant-Friedrichs-Lewy (CFL) condition where C_1 is imposed as 0.001:

$$\Delta t = C_1 = \frac{Re}{N^2} \tag{4}$$

The energy is determined as the following equation:

$$\frac{\partial_t E_k}{\delta t} = -\frac{2k^2}{Re}E_k - \left(\overline{\hat{u_k}} C_k\left(\hat{u_p}, \hat{u_q}\right) + \hat{u_k}\overline{C_k\left(\hat{u_p}, \hat{u_q}\right)}\right) + \overline{\hat{u_k}} F_k + \overline{F_k}\hat{u_k} \tag{5}$$

where $C_k\left(\hat{u_k}\hat{u_q}\right) \in C$ stands as the convective term. It should be acknowledged that it is calculated based on triadic interactions (see figure 1.

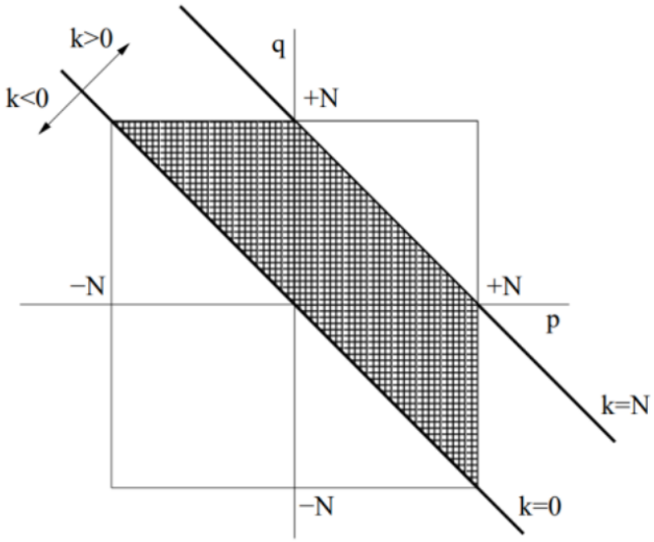


Figure 1. Triadic interactions

The non-linear convective term mostly facilitates the transfer of energy from low-frequency to high-frequency modes and vice versa in the transmission from large scales to small scales and from small scales to large scales.

For the diffusive term in the kinetic energy equation we have

$$-\frac{2k^2}{Re}E_k \tag{6}$$

This term reduces energy, especially impacting high-frequency patterns (small scales).

For the computation of the diffusive term, two methodologies have been used in order to compare their performance: DNS and LES.

Direct Numerical Solution (DNS) is based on the direct solution of the Navier Stokes equations without the need of any turbulence model, providing a fundamental method for the resolution of turbulent flows. The important term to remark is the viscosity, which is just based on the Reynolds number in this case as is exposed in equation 7

$$\nu = \frac{1}{Re} \tag{7}$$

Large-Eddy Simulation (LES) stands as a computationally efficient alternative to DNS, specially for coarse meshes. A spectral-eddy viscosity model is proposed in this simulation calculated as

$$\nu_t\left(\frac{k}{k_N}\right) = \nu + \nu_t^{\infty}\sqrt{\frac{E_k}{E_{k_N}}}\frac{1}{\sqrt{1+34.5e{-}3.03(k_N/k)}} \tag{8}$$

In equation 8, E_k represents the energy at the cut-off frequency k_N and $\nu_t^\infty = 0.31\frac{5-m}{m+1}\sqrt{\frac{3-m}{C_k}}$. Also the energy spectrum is assumed to have a fixed slope of m = 2 and $C_k = 0.4523$. Then the effect of the viscosity is given by $\nu_{eff}(k) = \nu + \nu_t(k)$.

2.1 Solver

The solver of the code is based on an iterative process that ends when the error between the new value of the variable and the one computed the iteration before is lower than a tolerance value fixed in this case at $1 \cdot 10^{-5}$. If this is not achieved, a maximum iteration number is also fixed at 10^5 iterations per loop.

The general algorithm follows this process:

1. Input data
2. Previous calculations
3. Main solver
4. Validations upload and result export

3 Code verification

Comparing the attained results with the provided plot serves as a means of validating this code for the study. Verification will occur in the following instances, as demanded by the assignment (see figure 2):

- Re = 40
- N (DNS) = 20
- N (DNS) = 100
- N (LES) = 20; $C_k = 0.4523$

In the upcoming part of the results 4, additional findings will be shown that compare various Reynolds values, Kolmogorov constants, and modes.

Figure 2 displays the results achieved using the code. It is evident that the graphs exhibit the anticipated form and mirror those put forth in the theory of the subject. Also, figure 3 validates the code for all simulations with N = 100.

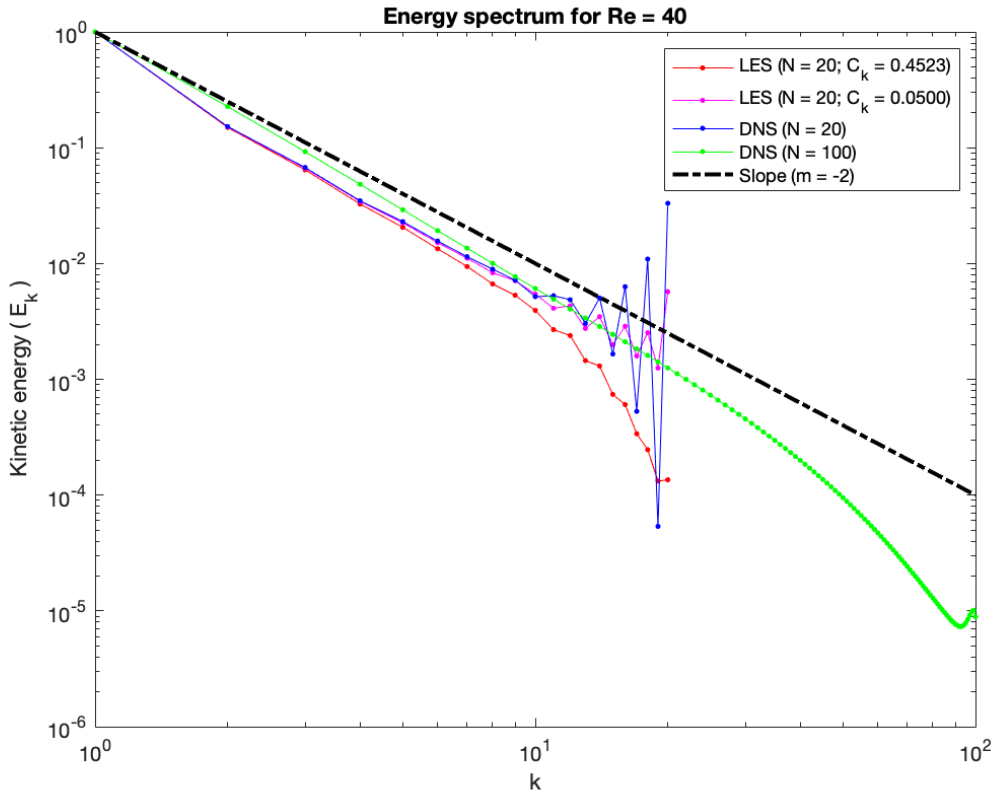


Figure 2. Verification: Energy spectrum for Re = 40 and N = 20

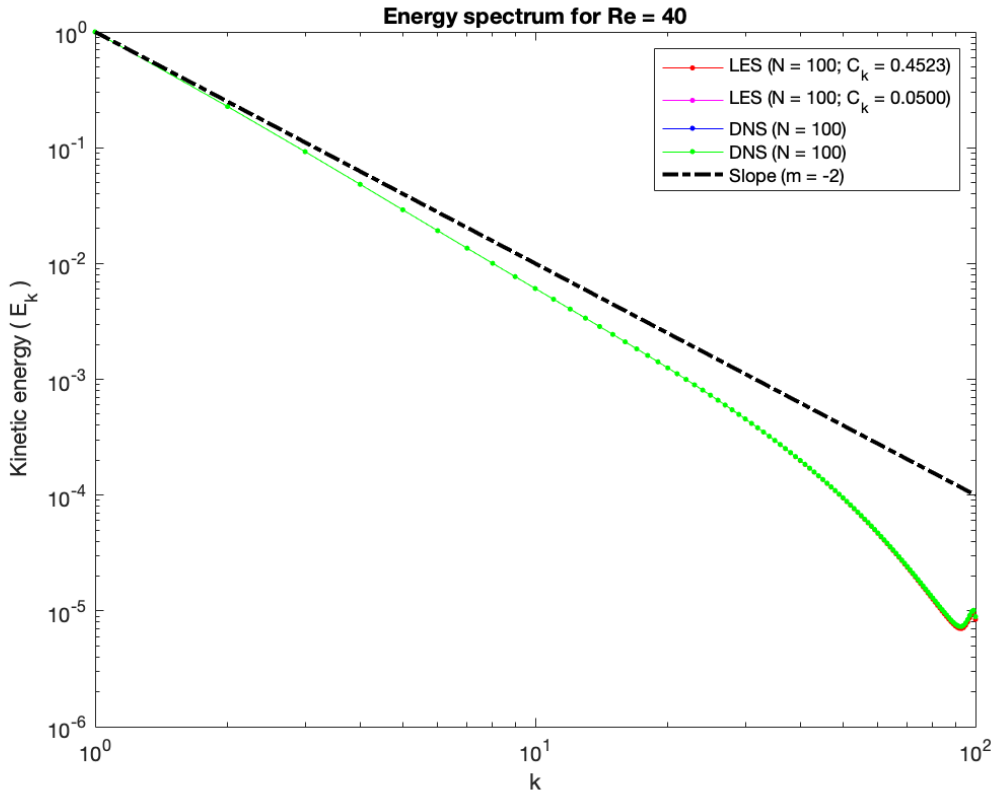


Figure 3. Verification: Energy spectrum for Re = 40 and N = 100

4 Results Analysis

Varying outcomes were achieved based on the choice between "Direct Numerical Simulation" (DNS) and Large-Eddy Simulation (LES).

Using DNS, simulations with 20 Fourier modes showed a failure to converge beyond k = 10, suggesting divergence for higher modes. The reason for this difference is due to too much energy being lost at small scales, causing the convective term to be overshadowed by the diffusive one. Additionally, utilising 100 Fourier modes resulted in a more positive result, as the curve closely mirrored the anticipated slope of m = 2. Nevertheless, the difference continued after k = 20, albeit not as pronounced as seen with fewer modes.

Using LES with a Kolmogorov constant of $C_k = 0.4523$ yielded outcomes comparable to DNS with 20 modes, but displayed increased viscosity at smaller scales, improving overall stability. The decrease in C_k starting at k = 10 led to improved outcomes due to reduced energy dissipation, with C_k falling to around one-tenth of its original value.

The plotted figures show the different simulations for the following Reynolds numbers: 20, 40, 80, 100, & 180. The higher the Reynolds number, the lower the accuracy for a fixed total number of modes N. Furthermore, simulations in DNS become more sensitive to this increase in Reynolds number, resulting in less accurate outcomes as the number rises.

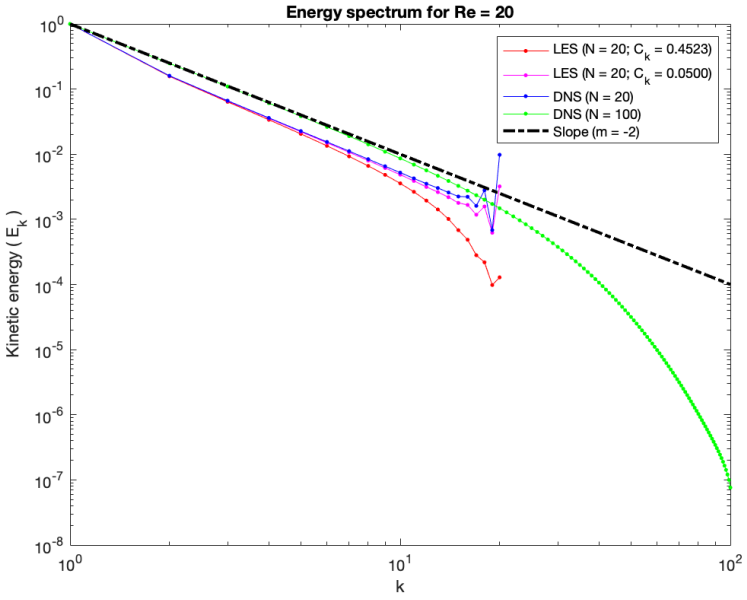


Figure 4. Energy spectrum for Re = 20

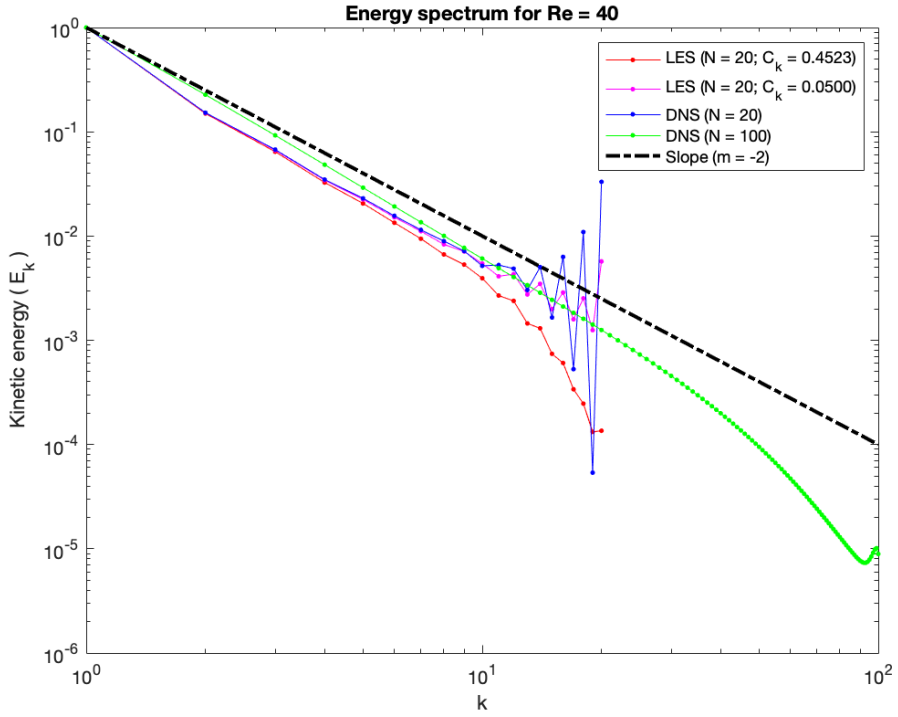


Figure 5. Energy spectrum for Re = 40

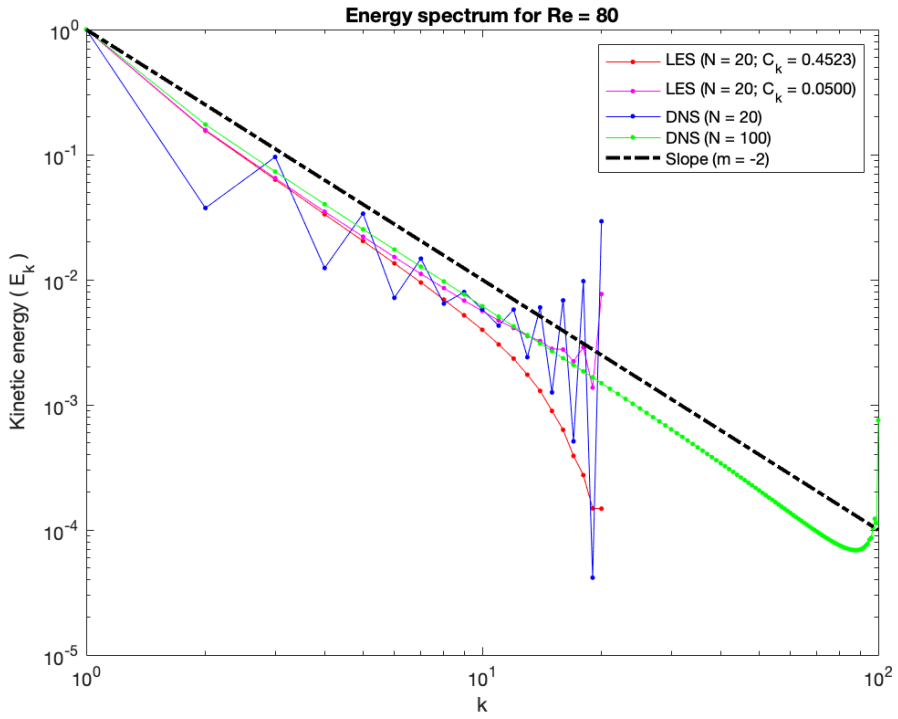


Figure 6. Energy spectrum for Re = 80

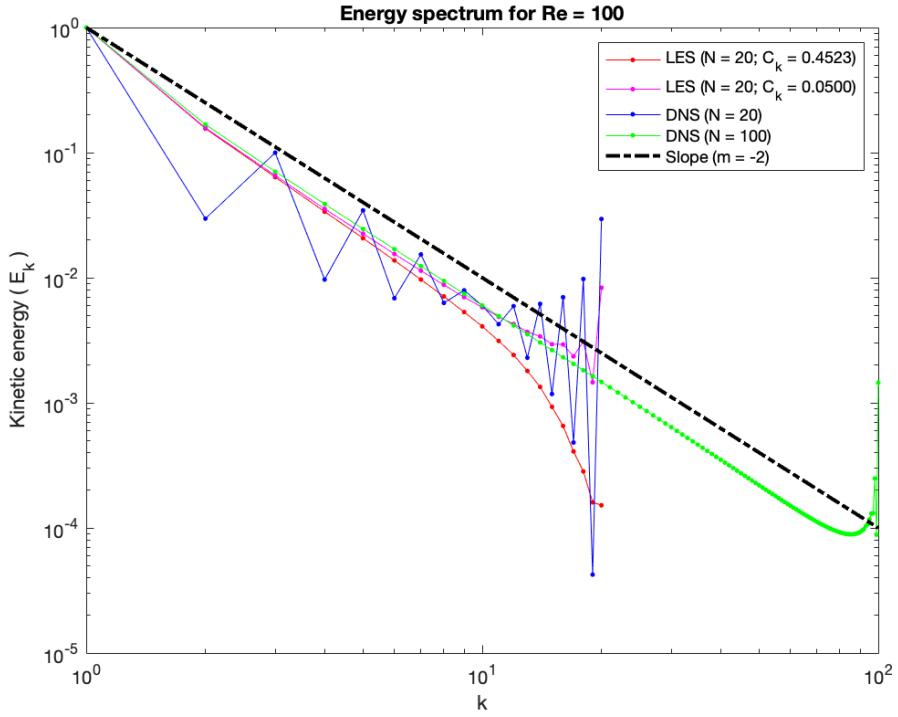


Figure 7. Energy spectrum for Re = 100

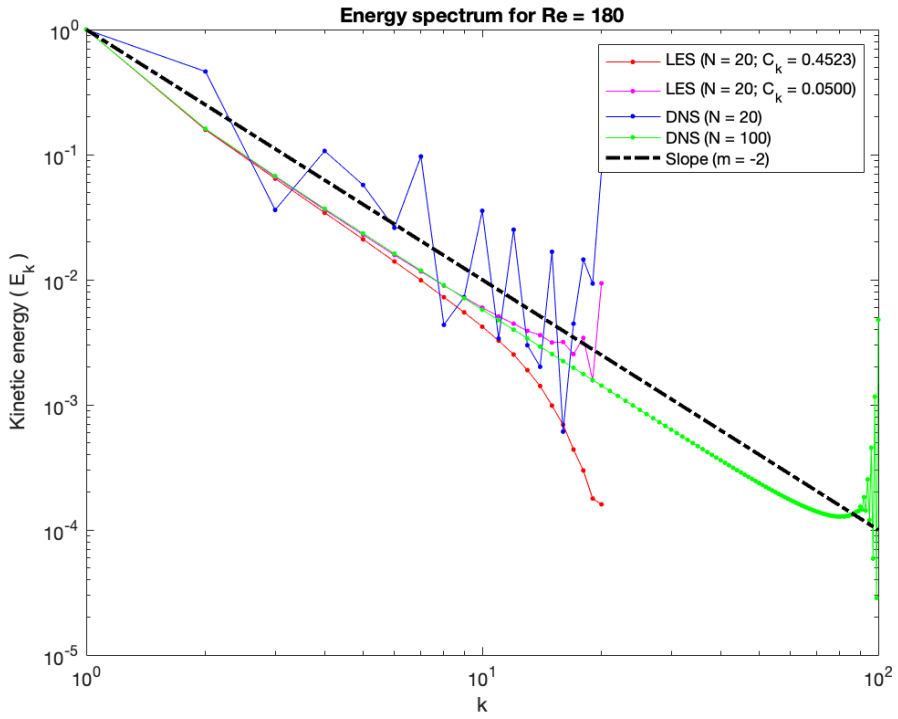


Figure 8. Energy spectrum for Re = 180

5 Conclusions

In conclusion, it can be said that the main goals of the task have been reached, with all suggested scenarios explored in the report and consistent results achieved. Therefore, the practical part of this task can be seen as finished.

As a way to inspire for this task, I want to mention that dealing with turbulence has been quite fascinating, as it is the least explored aspect of fluid dynamics during our time at university. After analysing the results, it is clear that there has been a thorough understanding of important factors related to convective and diffusive terms in turbulent fluids. Furthermore, researchers have investigated, utilised, and contrasted the Direct Numerical Simulation (DNS) and Large Eddy Simulation (LES) techniques. DNS focuses on smaller scales for better accuracy but at the cost of time efficiency, while LES prioritises larger scales for improved time efficiency and slightly lower accuracy.

Furthermore, it is crucial to recognise the difficult aspect of this task, especially because of the innovative use of turbulence solvers. However, this task has helped in gaining a more thorough comprehension of the theoretical principles of turbulence.

I want to end this last CFD assignment remarking the good experience this subject has been as a conclusion of my academic experience in fluid dynamics, englobing and closing all the aspects treated during the last 5 years.

A MATLAB Code

A.1 Main code

```
clear all;
close all;
clc;

% Author: Joel Campo
% 220311 - Computational Engineering. MUEA
% Assignment 4 (CFD): Turbulence

%% INPUT DATA:
% Physical data
Re = 40; % Reynolds number
C_k = 0.4523; % Kolmogorov values

% Other data
C1 = 0.001;
N = 20; % Fourier modes
m = 2; % Slope in the energy plot

% Performance values
tolerance = 1e-05;
maxIt = 10^5;

%% PREVIOUS CALCULATIONS
k = 1:N; % Mode number
u = 1./k; % Velocities
time_step = C1 * Re/(N^2);

E_k_store = zeros(1,N,3);

%% SOLVER
for case_study = 1:3 % 1 for DNS, 2+ for LES with
   combinations
    for i = 1:maxIt
        % Preallocate
        conv = zeros(1,N);
```

```
35          diff = zeros(1,N);
36
37          for j = 1:N
38              % Convective term
39              for p = (-N+j):N
40                  q = j-p;
41                  if p~=0 && q~=0
42                      if p > 0
43                          u_p = u(p);
44                      else
45                          u_p = conj(u(-p));
46                      end
47
48                      if q > 0
49                          u_q = u(q);
50                      else
51                          u_q = conj(u(-q));
52                      end
53                      conv(j) = conv(j) + u_p * q * 1i *
    u_q;
54                  end
55              end
56
57              % Diffusive term
58              if case_study == 1 % DNS
59                  diff(j) = (k(j)^2) * u(j) / Re;
60
61              elseif case_study == 2 % LES
62                  MetLes = 0.31 * (5 - m)/(m + 1) * sqrt(3
    - m) * C_k^(-3/2);
63                  eddy = 1 + 34.5 * exp(-3.03 * (k(N)/k(j))
    );
64                  E_k_n = u(N) * conj(u(N));
65                  nu = MetLes * (E_k_n / k(N))^0.5 * eddy;
66
67                  diff(j) = k(j)^2 * u(j) * (1/Re + nu);
68
69              elseif case_study == 3 % LES C_k = 0.05
70                  C_k = 0.05;
```

```
71                  MetLes = 0.31 * (5 - m)/(m + 1) * sqrt(3
   - m) * C_k^(-3/2);
72                  eddy = 1 + 34.5 * exp(-3.03 * (k(N)/k(j))
   );
73                  E_k_n = u(N) * conj(u(N));
74                  nu = MetLes * (E_k_n / k(N))^0.5 * eddy;
75
76                  diff(j) = k(j)^2 * u(j) * (1/Re + nu);
77              else
78                  error('The case study does not exist');
79              end
80          end
81
82          % Velocities
83          u_after = u + time_step * (conv - diff);
84          u_after(1) = 1;
85
86          % Error
87          err = abs(u_after - u);
88          if max(err(:)) < tolerance
89              break
90          else
91              u = u_after;
92          end
93      end
94
95      % Kinetic energy
96      E_k = u_after.*conj(u_after);
97      E_k_store(:,:,case_study) = E_k;
98  end
99
100 if Re == 20
101     DNS_load = load("DNS20.mat");
102 elseif Re == 40
103     DNS_load = load("DNS40.mat");
104 elseif Re == 80
105     DNS_load = load("DNS80.mat");
106 elseif Re == 100
107     DNS_load = load("DNS100.mat");
```

```
elseif Re == 180
    DNS_load = load("DNS180.mat");
else
    error("DNS(N=100) not computed for this Reynolds number");
end

figure(1)
loglog (k,E_k_store(:,:,3),'r','Marker','.')
hold on
loglog (k,E_k_store(:,:,2),'m','Marker','.')
hold on
loglog (k,E_k_store(:,:,1),'b','Marker','.')
hold on
k = 1:100;
loglog (k,DNS_load.E_k,'g','Marker','.')
hold on
loglog (k,1./k.^m, 'k-.', 'LineWidth', 2)
legend ('LES (N = 20; C_k = 0.4523)','LES (N = 20; C_k = 0.0500)', ...
    'DNS (N = 20)','DNS (N = 100)','Slope (m = -2)')
xlabel('k')
ylabel('Kinetic energy ( E_k )')
title('Energy spectrum for Re = 100')
```